

CDS CURVES

Global Credit Derivative and Index Research

Dmytro Shelukhin ^{AC}

Credit Index & Derivative Strategy

J.P. Morgan Securities Plc

dmytro.shelukhin@jpmorgan.com

See the end pages of this presentation for analyst certification and important disclosures.

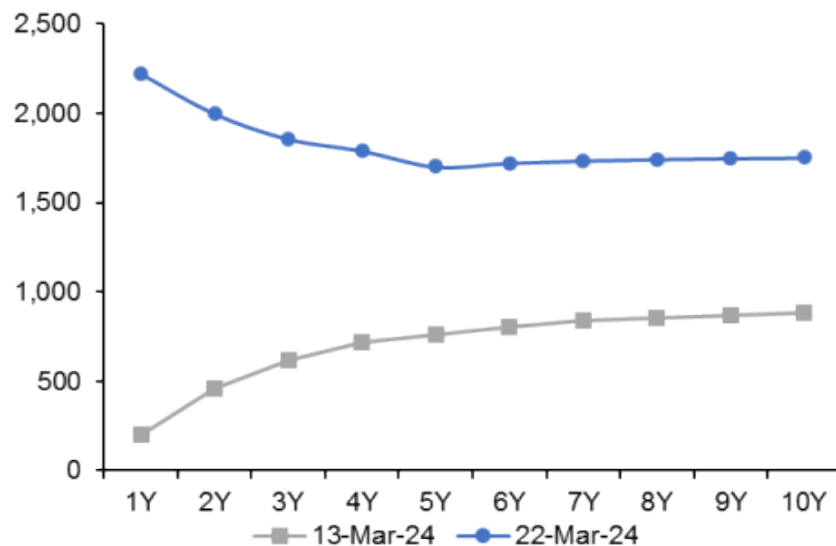
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CDS Curves: Credit Risk for Different Maturities

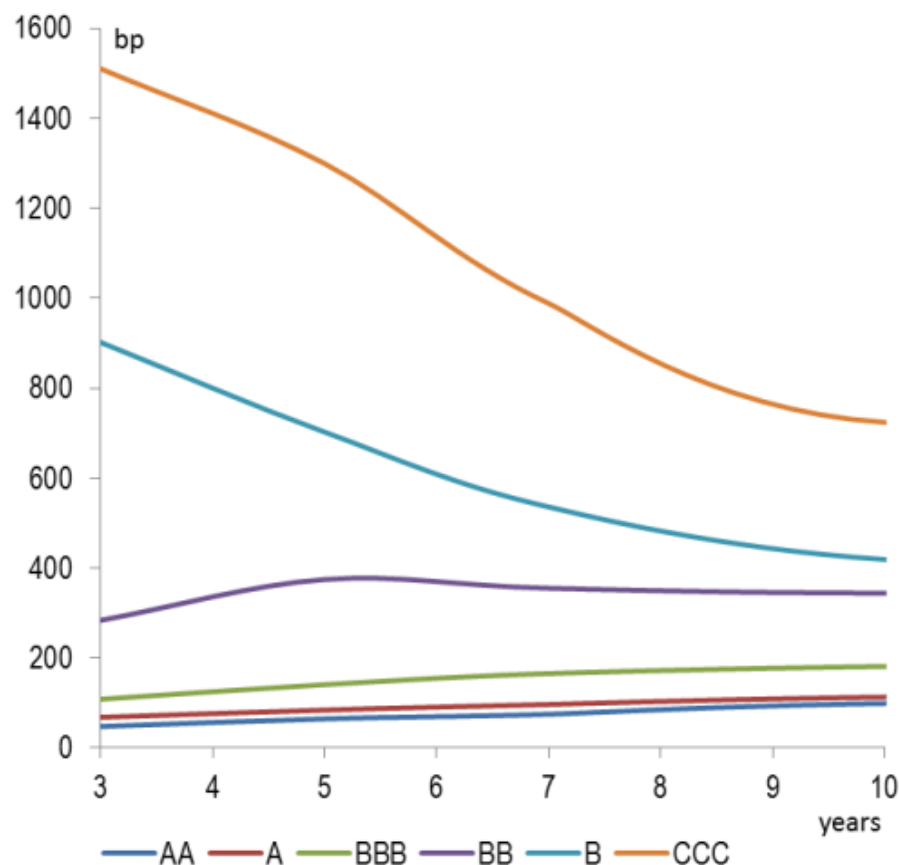
- Credit default swaps on the same company can have different spreads for different maturities.
- The entire term structure of CDS spreads is known as the “CDS Curve”.
- Depending on the environment/situation, CDS curves can be upward/downward sloping, humped...
- Example: Altice France (SFRFP)



Source: J. P. Morgan

Curves Steepness and Credit Quality

Spread Curves vs. Rating Category



Source: J.P. Morgan

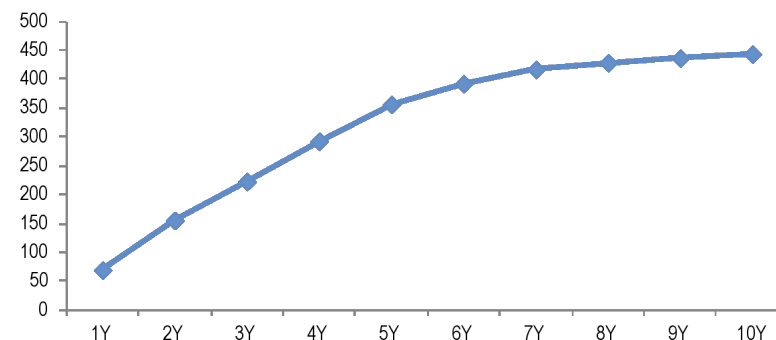
- General empirical rule:
 - Credit curves of high-rated companies tend to be upward sloping
 - While curves of low-rated companies tend to be downward sloping
- Intuition:
 - High-rated companies are relatively secure in the short term, but uncertainty increases longer term (e.g., might be downgraded from HG to HY)
 - Low-rated companies are very risky in the short term, but situation might improve if short-term hurdle is passed
- CDS curves give us information about how the default risk of a company is expected to change over time.

CDS Curve Behaviour – Tight Spread Names

Uncertainty premium

- For a “normal” name, it is common to see an upward sloping CDS curve.
- A lot of this is due to the uncertainty premium for longer dated contracts.
- There is a lot of transparency over a company’s finances in the short term, but for longer contracts there is a lot less visibility.
- Investors typically demand an “uncertainty” premium for selling protection on longer dated contracts as a result of this.
- This causes curves to be upward sloping in benign environments.

Upward sloping curves



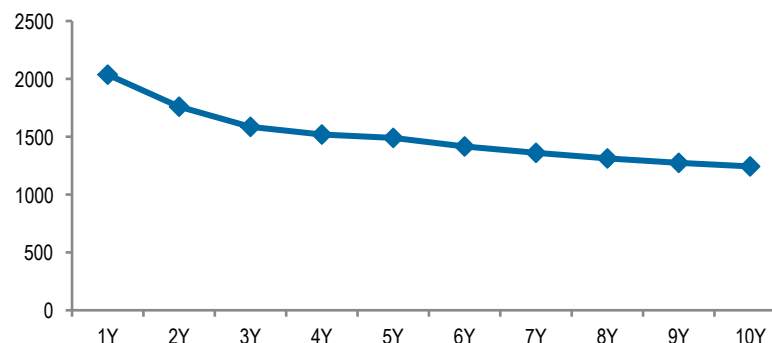
Source: J.P. Morgan

CDS Curve Behaviour – Distressed Names

Uncertainty premium

- When names are close to default, the upfronts of all contracts trade close to $(1 - \text{Recovery})$ independent of maturity.
- This causes the CDS curve to invert (in spread terms).
- From a default probability standpoint this is reflective of the front-loaded default risk in the name.
- During periods of large macro stress, the entire index curve can invert.

Inverted curves



Source: J.P. Morgan

Upfront Costs for Distressed Name

1Y	48 $\frac{1}{2}$	... 51 $\frac{1}{2}$
2Y	48 $\frac{3}{4}$	... 51 $\frac{3}{4}$
3Y	49	... 52
4Y	49 $\frac{1}{2}$	... 52 $\frac{1}{2}$
5Y	50 $\frac{1}{2}$	... 52 $\frac{1}{2}$
7Y	50 $\frac{1}{2}$	... 52 $\frac{1}{2}$
10Y	50 $\frac{1}{2}$	... 52 $\frac{1}{2}$

Source: J. P. Morgan, Bloomberg

Trading CDS Curves

- Why would we want to take a view on CDS curves?
- CDS curve trades allow you to take a view on the timing of defaults, rather than the outright likelihood of defaults happening.
- There are two main ways that people play curve trades:
 - Equal notional trades
 - Duration weighted trades
- Terminology:
 - Buy long dated protection, sell short dated protection: **Steepener**
 - Sell long dated protection, buy short dated protection: **Flattener**
- We can calculate the spread exposure of a curve trade using the durations of each leg.
- P&L for a 3y vs. 5y curve flattener based on spread changes:

$$P\&L = \Delta S_{3Y} \times Notional_{3Y} \times Duration_{3Y} - \Delta S_{5Y} \times Notional_{5Y} \times Duration_{5Y}$$

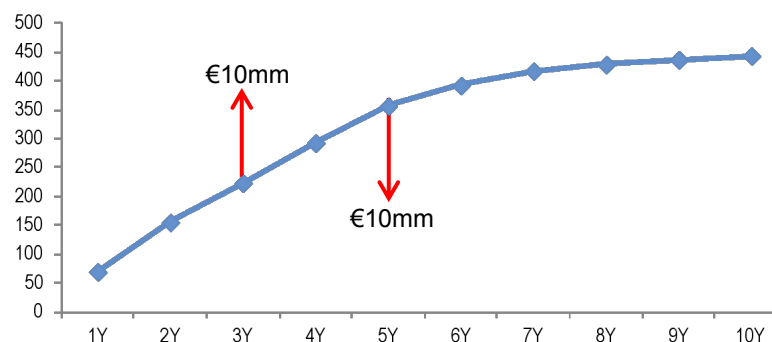
Equal Notional Trades

Equal Notional Curve Trades

- If we sell 5y CDS protection we will have default exposure to the company for the next 5 years.
- Let's say we only want to have exposure for years 4 and 5, because we are concerned about default risk prior to that.
- We can buy 3y protection and sell 5y protection on an equal notional.
- If the company defaults before 3y, the default payouts from our 3y and 5y legs will cancel out.
- This is known as an **3s5s Equal Notional Flattener**.
- This trade will perform well if the 3y widens compared to the 5y.
- However, because the duration of the 5y leg is higher than the 3y leg, this trade also has exposure to parallel moves in spreads.
- An equal notional flattener will lose money if spreads widen in parallel.
- **Equal notional curve trades have both curve and spread risk.**

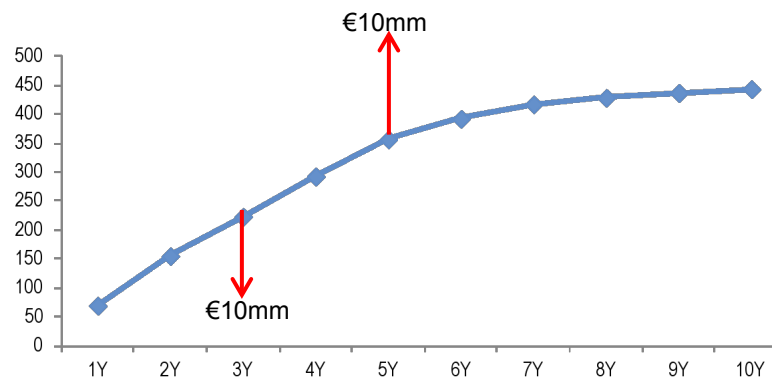
Source: J.P. Morgan

3s5s Equal Notional Flattener



Source: J. P. Morgan.

3s5s Equal Notional Steepener



Source: J. P. Morgan.

Equal Notional Curve Trades - continued

- Equal notional curve trades are effectively buying or selling forward-starting CDS protection.
- For example, if I enter a 3s5s equal notional flattener (buy 3y protection, sell 5y protection in equal notionals) I am effectively selling 2y protection starting in 3 years.
- Forward starting contracts do not trade by themselves in the market; you have to construct them with an equal notional curve trade.
- We can calculate the implied spread for a forward starting contract using the spreads and durations of the two contracts we use to construct the equal notional trade.

$$\text{Implied Forward Spread} = \frac{\text{Spread}_2 \times \text{Duration}_2 - \text{Spread}_1 \times \text{Duration}_1}{\text{Duration}_2 - \text{Duration}_1}$$

- For example:

3y Spread = 100 Duration = 2.7

5y Spread = 150 Duration = 4.5

Implied Forward Spread between 3 and 5y = $(150 \times 4.5 - 100 \times 2.7) / (4.5 - 2.7) = 225\text{bp}$

- Steep curves imply high forwards.

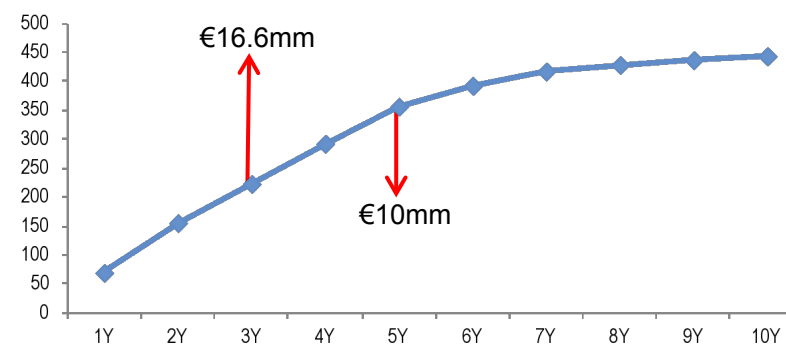
Duration-Weighted Trades

Duration-Weighted Trades

- If we want our trade to have no exposure to parallel spread moves, we need to change the notional size of each leg to account for the different durations of each leg.
- Example:
3s5s Flatteners: Sell 5y protection, buy 3y protection.
3y duration = 2.7
5y duration = 4.5
5y notional = €10,000,000
- To be neutral to parallel shifts in the curve we need to overweight the notional of the 3y leg to account for the lower duration.
- 3y notional = $5y \text{ duration} / 3y \text{ duration} \times 5y \text{ notional}$
= $4.5 / 2.7 \times €10,000,000 = €16,666,666$.
- If 5y widens by 1bp, we make $4.5 \times 1bp \times €10mm = €4,500$.
- If 3y widens by 1bp, we lose $2.7 \times 1bp \times €16.7mm = €4,500$
- **Duration-weighted curve trades only have exposure to changes in the steepness of the curve, not to outright spread moves.**

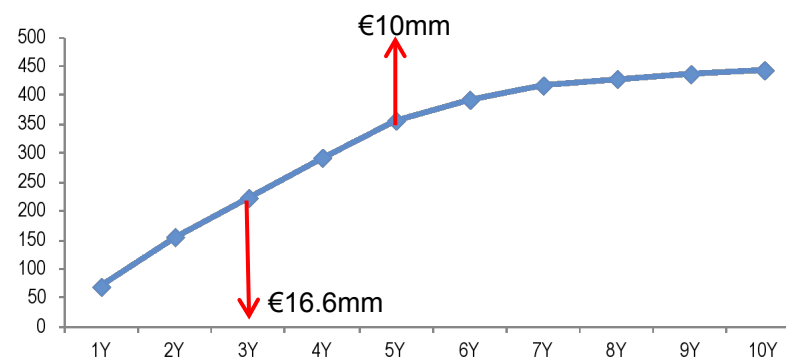
Source: J.P. Morgan.

3s5s Duration Weighted Flatteners



Source: J.P. Morgan

3s5s Duration Weighted Steepener



Source: J. P. Morgan.

Duration-Weighted Trades - continued

- Duration-weighted curve trades are used to take pure exposure to curve shapes and the timing of defaults.
- If we think default risk is going to be come more front loaded (curve flattens/inverts) we should enter a duration-weighted flattener.
- If we think default risk is going to become more back-dated we should enter a duration-weighted steepener.
- Can be used to play views on debt maturities, refinancing, short-term default risks, etc. without taking large mark-to-market volatility associated with buying or selling outright CDS protection.

Curve Trade P&L

Spread Exposure

- When looking at the sensitivities of curve trades we look at both the exposure to spread changes and how the P&L changes with time.
- For spread exposure, P&L for a flattener is given by:

$$P\&L = \Delta S_{3Y} \times Notional_{3Y} \times Duration_{3Y} - \Delta S_{5Y} \times Notional_{5Y} \times Duration_{5Y}$$

- For Equal Notional Flattener, we can split this P&L up into curve risk and spread risk:

$$P\&L = Notional \times \left[\underbrace{(\Delta S_{3y} - \Delta S_{5y}) \times Dur_{5y}}_{\text{Curve Risk}} - \underbrace{(Dur_5 - Dur_3) \times \Delta S_{3y}}_{\text{Spread Risk}} \right]$$

- For Duration-Weighted Flattener, we only have curve risk (given $Notional_{3y} = Dur_{5y} / Dur_{3y} \times Notional_{5y}$).

$$P\&L = Notional_{5y} \times \underbrace{(\Delta S_{3y} - \Delta S_{5y}) \times Dur_{5y}}_{\text{Curve Risk}}$$

Example P&L Question 1

Spread Exposure

1. We enter a 5s10s equal notional steepener on ABC Corp with the following details.

5y Spread = 100bp

5y Duration = 4.5

10y Spread = 120bp

10y Duration = 8.5

Notional = €10,000,000

The 5y spread widens to 115bp and the 10y spread widens to 130bp immediately after we enter the trade.

a) What is the P&L?

We are selling 5y protection and buying 10y protection.

5y CDS P&L = $-4.5 \times 15\text{bp} \times €10,000,000 = -€67,500$

10y CDS P&L = $8.5 \times 10\text{bp} \times €10,000,000 = €85,000$

Total = €17,500 Profit

b) What is the implied CDS 5y10y forward before and after the spread move?

Before move:

Forward = $(120 \times 8.5 - 100 \times 4.5) / (8.5 - 4.5) = 142\text{bp}$

After move:

Forward = $(130 \times 8.5 - 115 \times 4.5) / (8.5 - 4.5) = 146.875\text{bp}$

Source: J.P. Morgan

Example P&L Question 2

Spread Exposure

1. We enter a 3s5s duration weighted flattener on XYZ Corp with the following details:

3y Spread = 350bp 3y Duration = 2.5

5y Spread = 450bp 5y Duration = 4.0

5y Notional = €10,000,000

a) What notional of 3y will we trade?

$$3y \text{ Notional} = 5y \text{ Duration} / 3y \text{ Duration} \times 5y \text{ Notional} = 4 / 2.5 \times €10,000,000 = €16,000,000$$

b) The curve steepens from 100bp to 150bp. What is our P&L?

$$\begin{aligned} \text{P\&L for duration weighted flattener} &= 5y \text{ Notional} \times 5y \text{ Duration} \times \text{Curve Flattening} \\ &= €10,000,000 \times 4 \times -50\text{bp} \\ &= \text{Loss of } €200,000 \end{aligned}$$

Source: J.P. Morgan

Time Value: Carry

Carry

- What is the carry on curve trades?
- In the simplest form, we can look at the spread of each leg.
- For a flattener trade, annual carry is equal to:

$$\text{Carry} = S_{5y} \times \text{Notional}_{5y} - S_{3y} \times \text{Notional}_{3y}$$

- Carry for duration-weighted flatteners is usually negative, except for very steep curves.
- Carry for equal notional flatteners is positive for most curves.
- In reality, CDS trades on a fixed coupon plus upfront basis, not full running spread.
- The actual “carry” we get each year is from the fixed coupons of each leg, and there is also some change in the upfronts due to the passage of time.
- Despite this, the full running spread is a good approximation for the carry.

Source: J.P. Morgan

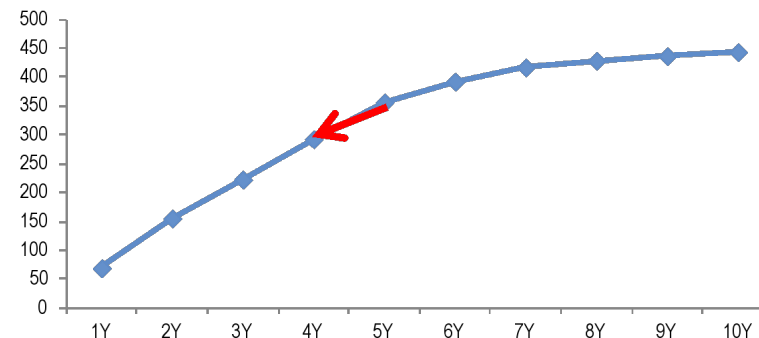
Time Value: Slide I

Slide

- Let's say we sell 5y protection today at 350bp and carry out a scenario analysis of this trade over the next year.
- In one year's time this contract will be a 4y trade. We need to make an assumption about where the 4y contract will trade in 1y.
- The assumption that most people in the market use is that one year from now, a 5y contract today will trade with a spread equal to that of the current 4y contract.
- This effect is called “slide”; we expect spreads to “slide” down the CDS curve as time passes.
- When we consider the P&L from curve trades as time passes we should take this slide into account.
- Importantly though, slide is not always realised. It's a “best guess” at what credit curves in the future will look like.

Source: J.P. Morgan

Slide from 5y to 4y



Time Value: Slide II

Slide

- We can estimate the slide of a single CDS contract by looking at the current spread curve as well as the duration in one year's time.
- The slide for a seller of protection is equal to:

$$\text{Slide} = (\text{Spread today} - \text{Spread in 1y}) \times \text{Duration in 1y} \times \text{Notional}$$

- For example, we sell 5y protection at 350bp today. The current 4y spread is 300bp and has a duration of 3.6. The notional is €10mm.
 - $\text{Slide} = (350\text{bp} - 300\text{bp}) \times 3.6 \times €10,000,000 = €180,000$
- To calculate the entire value of a position over one year, we need to combine the slide and carry.
- **Time Value = Carry + Slide**

Source: J.P. Morgan

Example Time Value Questions

Time value of outright position

Carry and Slide

1. We buy protection ABC Corp 5y protection for 200bp on a notional of €10,000,000

Current 4y spread = 180bp

4y duration = 3.6

5y duration = 4.5

a) What is the carry on this position for one year?

$$\text{Carry} = -200\text{bp} \times 1 \text{ year} \times €10,000,000 = -€200,000$$

b) What is the slide?

$$\begin{aligned} \text{Slide} &= - (5\text{y Spread} - 4\text{y Spread}) \times \text{Duration in 1y} \times \text{Notional} \\ &= - (200\text{bp} - 180\text{bp}) \times 3.6 \times €10,000,000 \\ &= - €72,000 \end{aligned}$$

c) What is the time value?

$$\text{Time Value} = \text{Carry} + \text{Slide} = -200,000 + (-72,000) = -€272,000$$

Source: J.P. Morgan

Example Time Value Questions

Time value of equal notional steepener steepener

Carry and Slide

1. We enter a 3s5s equal notional steepener on ABC Corp on a notional of €10,000,000

Tenor	Spread	Duration
1Y	120	0.9
2Y	140	1.8
3Y	160	2.7
4Y	180	3.6
5Y	200	4.5

a) What is the time value of this position?

We are buying €10,000,000 of 5y protection and selling €10,000,000 of 3y protection. The 5y leg is same as previous example, which had a time value of -€272,000.

For the 3y leg, carry is equal to $160\text{bp} \times 1 \text{ year} \times €10,000,000 = €160,000$

Slide is equal to $(160\text{bp} - 140\text{bp}) \times 1.8 \times €10,000,000 = €36,000$

Time Value = +€196,000 for 3y leg

Total Time Value = Time Value for 3y Leg + Time Value for 5y Leg = $€196,000 - €272,000 = -€76,000$

Source: J.P. Morgan

Default Exposure

Impact of Defaults on Curve Trades

- Curve trades can have varying types of jump to default exposure. The profit from defaults for a 3s5s flattener is equal to:

$$DefaultExposure = Notional_{3y}(1 - R_{3y}) - Notional_{5y}(1 - R_{5y})$$

- **Duration-weighted flatteners** usually have a **positive jump-to-default** because the short-dated notional will be higher than the long-dated notional.
- Equal notional trades usually have minimal default exposure due to two legs cancelling each other out.
- The exact default exposure will depend on the relative recovery rates for the two legs.
- For Bankruptcy, Failure to Pay and Governmental Intervention credit events the recoveries will be equal for all maturities.
- For Restructuring events shorter-dated contracts may have higher recoveries.

Additional Material

Additional Material

- *Credit Derivatives Handbook*, 2006, J.P. Morgan
- *Trading Credit Curves I*, 2006, J.P. Morgan
- *Trading Credit Curves II*, 2006, J.P. Morgan
- *Trading CDS with Fixed Coupons: How it affects basis and curve positions*, 2010, J.P. Morgan.

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CDS Curves

CDS Curves

- We can calculate a probability of default from the CDS spread.
- **Average annual probability of default** = CDS Spread / (1-Recovery).
- For a name trading with a 5y spread of 200bp and R=40%, the **average annual probability of default** is $\lambda_5 = 3.33\%$.

Average annual probability of default over i years = λ_i

- This is an average over the 5y of the contract.
- Hazard rates are **conditional probabilities of default**; i.e. what is the probability that a company defaults in year 3, given that it has not defaulted in years 1 or 2?

ith year Hazard Rate = Probability of Defaulting in Year i given survival up until Year i-1 = $\lambda_i | i-1$

- Some companies are very safe for the next few years and very risky after that – simply looking at the 5y spread will not tell you this.
- **CDS curves give us information about how the default risk of a company is expected to change over time.**

Source: J.P. Morgan

Hazard Rates

Hazard Rates

- Hazard rates are **conditional probabilities of default**; i.e. what is the probability that a company defaults in year 3, given that it has not defaulted in years 1 or 2?
- We can calculate hazard rates for a particular year using the CDS curve. We assume a recovery of 40%.
- i.e. **Company ABC**
1 Year CDS Spread = 100bp Average annual probability of default = 1.66% (=100bp/(1-0.4))
2 Year CDS Spread = 200bp Average annual probability of default = 3.33% (= 200bp/(1-0.4))
- If the probability of default over the first year is 1.66%, but the average annual probability of default over the first two years is 3.33%, then intuitively the probability of default in the second year must be higher than that in the first year (in order to make the average higher).
- There are two ways we can calculate the **probability of survival** at the end of the second year:
 - $PS_2 = (1 - 3.33\%)(1 - 3.33\%)$ i.e. the probability of survival based on the average default probability we obtain from the 2 year CDS contract.
 - $PS_2 = (1 - 1.66\%)(1 - \lambda_{2|1})$ i.e. the probability of surviving the first year (based on the 1y CDS spread) multiplied by the (unknown) conditional probability of survival for the second year.
- λ_2 is the hazard rate for the second year.
- By equating these two expressions, we can obtain an expression for the second year hazard rate.

$$(1 - 3.33\%)^2 = (1 - 1.66\%)(1 - \lambda_{2|1})$$

$$\lambda_{2|1} = 1 - \frac{(1 - 3.33\%)^2}{1 - 1.66\%} = 4.97\%$$

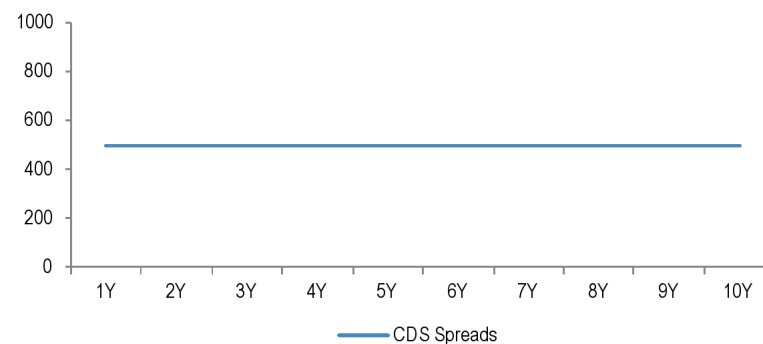
Source: J.P. Morgan

Flat Curves

Flat CDS Curves

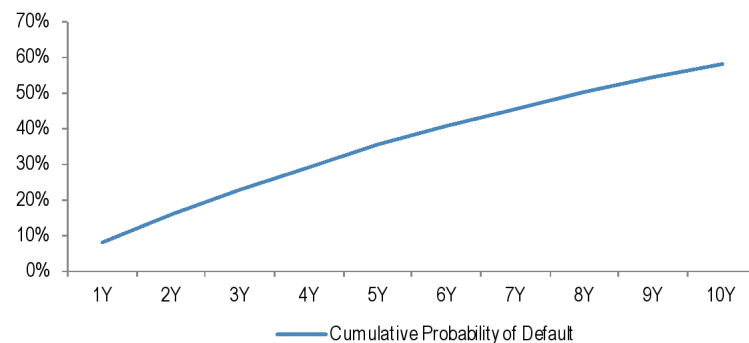
- Names with flat curves also have flat hazard rate curves.
- The riskiness of the name is not changing with time – it has an equal probability of defaulting in each of the next 10 years.
- The cumulative probability of default is still increasing however.

CDS Spread Curves



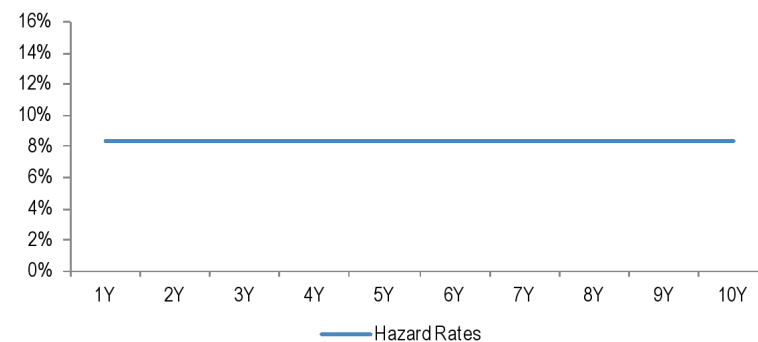
Source: J.P. Morgan

Cumulative Probabilities of Default



Source: J.P. Morgan

Hazard Rates



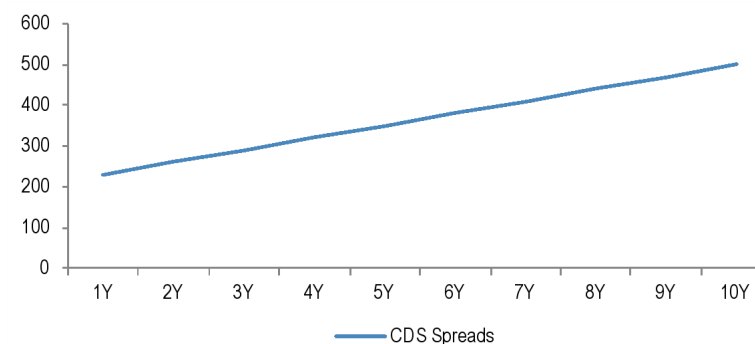
Source: J.P. Morgan

Upward Sloping Curves

Upward Sloping CDS Curves

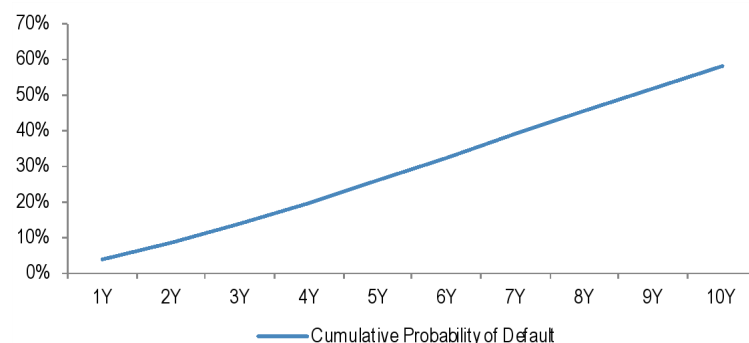
- Names with upward sloping (or steep) curves have increasing hazard rates.
- Each year is riskier than the last.
- The cumulative probability of default is always increasing, but more slowly at first than the flat curve and more quickly at the end.

CDS Spread Curves



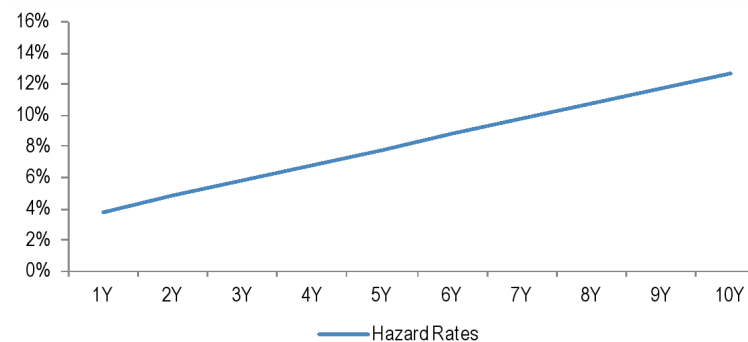
Source: J.P. Morgan

Cumulative Probabilities of Default



Source: J.P. Morgan

Hazard Rates



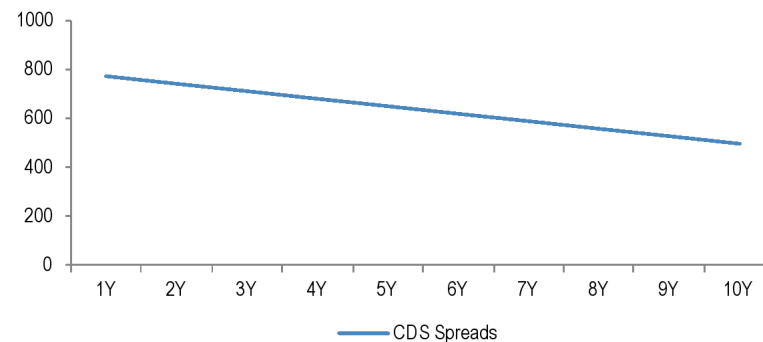
Source: J.P. Morgan

Downward Sloping Curves

Downward Sloping CDS Curves

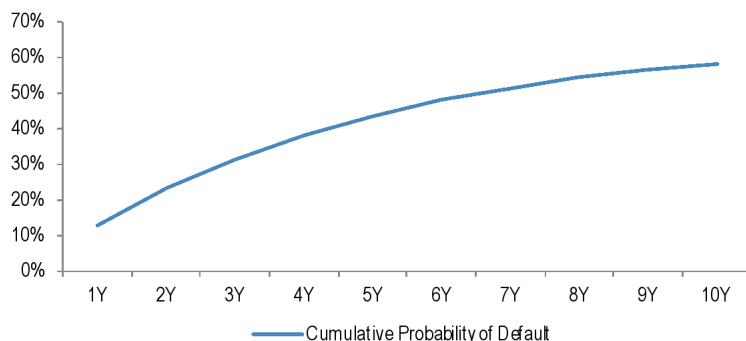
- Names with downward sloping (or inverted) curves have decreasing hazard rates
- The first few years are the most risky, with annual default risk falling over time.
- The cumulative probability of default is always increasing, but more quickly at first than the flat curve and more slowly at the end.
- Names with inverted curves are often close to default.

CDS Spread Curves



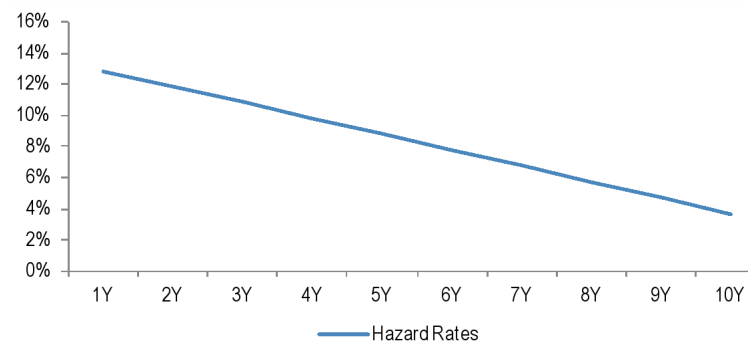
Source: J.P. Morgan

Cumulative Probabilities of Default



Source: J.P. Morgan

Hazard Rates



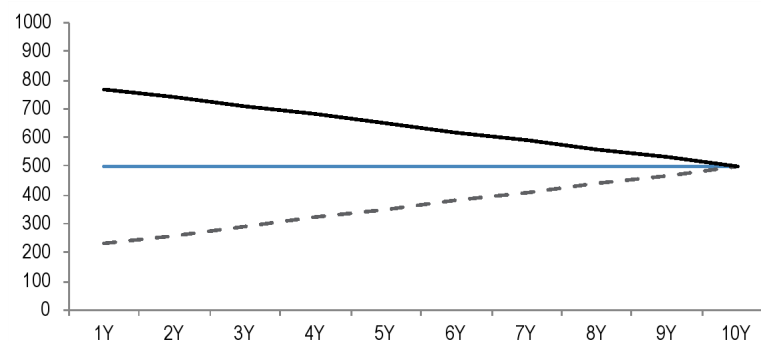
Source: J.P. Morgan

Comparison

Comparison of different curves

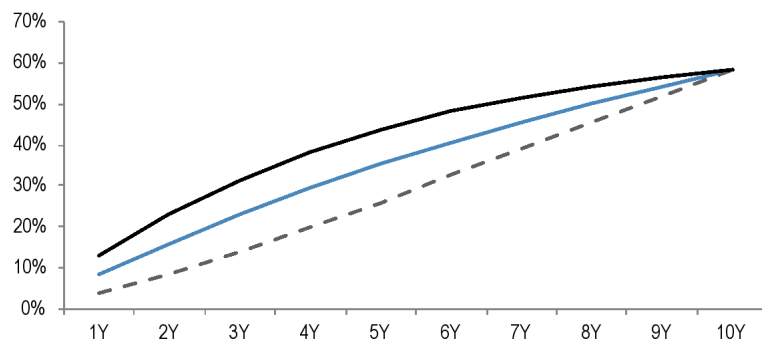
- The 10y spread of each curve is the same.
- This means the total amount of default risk over the whole 10 years is the same for each curve.
- This can be seen from the cumulative probabilities of default which converge at the 10y point.
- The hazard rates are very different though; the default risk is distributed very differently through time.

CDS Spread Curves



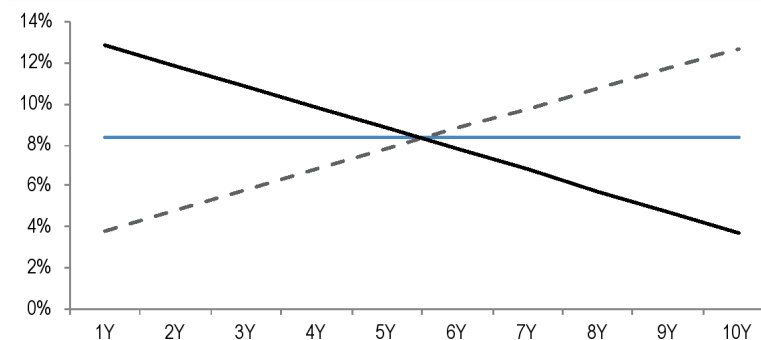
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Cumulative Probabilities of Default



Source: J.P. Morgan

Hazard Rates



Source: J.P. Morgan

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Disclosures

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